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Equity Valuation: Concepts and Basic Tools

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MULTISTAGE DIVIDEND DISCOUNT MODELS

Learning Outcomes

- calculate and interpret the intrinsic value of an equity security based on the Gordon (constant) growth dividend discount model or a two-stage dividend discount model, as appropriate
- identify characteristics of companies for which the constant growth or a multistage dividend discount model is appropriate
- explain advantages and disadvantages of each category of valuation model

Multistage growth models are often used to model rapidly growing companies. The *two-stage DDM* assumes that at some point the company will begin to pay dividends that grow at a constant rate, but prior to that time the company will pay dividends that are growing at a higher rate than can be sustained in the long run. That is, the company is assumed to experience an initial, finite period of high growth, perhaps prior to the entry of competitors, followed by an infinite period of sustainable growth. The two-stage DDM thus makes use of two growth rates: a high growth rate for an initial, finite period followed by a lower, sustainable growth rate into perpetuity. The Gordon growth model is used to estimate a terminal value at time n that reflects the present value at time n of the dividends received during the sustainable growth period.

Equation 11 will be used here as the starting point for a two-stage valuation model. The two-stage valuation model is similar to Example 8 except that instead of assuming zero dividends for the initial period, the analyst assumes that dividends will exhibit a high rate of growth during the initial period. Equation 11 values the dividends over the short-term period of high growth and the terminal value at the end of the period of high growth. The short-term growth rate, g_S, lasts for n years. The intrinsic value per share in year n, V_n, represents the year n value of the dividends received during the sustainable growth period or the terminal value at time n. V_n can be estimated by using the Gordon growth model as shown in Equation 12, where g_L is the long-term or sustainable growth rate. The dividend in year $n + 1$, D_{n+1}, can be determined by using Equation 13:

$$V_0 = \sum_{t=1}^{n} \frac{D_0(1+g_S)^t}{(1+r)^t} + \frac{V_n}{(1+r)^n} \qquad 11$$

$$V_n = \frac{D_{n+1}}{r - g_L} \qquad 12$$

$$D_{n+1} = D_0(1+g_S)^n(1+g_L) \qquad 13$$

EXAMPLE 9

Applying the Two-Stage Dividend Discount Model

1. The current dividend, D_0, is \$5.00. Growth is expected to be 10 percent a year for three years and then 5 percent thereafter. The required rate of return is 15 percent. Estimate the intrinsic value.

Hide Solution

Solution:

D_1 = \$5.00(1 + 0.10) = \$5.50

D_2 = \$5.00$(1 + 0.10)^2$ = \$6.05

D_3 = \$5.00$(1 + 0.10)^3$ = \$6.655

$$V_0 = \frac{\$5.50}{(1+0.15)} + \frac{\$6.05}{(1+0.15)^2} + \frac{\$6.655}{(1+0.15)^3} + \frac{\$54.8775}{(1+0.15)^3} \approx \$59.68$$

The DDM can be extended to as many stages as deemed appropriate. For most publicly traded companies (that is, companies beyond the start-up stage), practitioners assume growth will ultimately fall into three stages: 1) growth, 2) transition, and 3) maturity. This assumption supports the use of a *three-stage DDM*, which makes use of three growth rates: a high growth rate for an initial finite period, followed by a lower growth rate for a finite second period, followed by a lower, sustainable growth rate into perpetuity.

One can make the case that a three-stage DDM would be most appropriate for a fairly young company, one that is just entering the growth phase. The two-stage DDM would be appropriate to estimate the value of an older company that has already moved through its growth phase and is currently in the transition phase (a period with a higher growth rate than the sustainable growth rate) prior to moving to the maturity phase (the period with a lower, sustainable growth rate).

However, the choice of a two-stage DDM need not rely solely on the age of a company. Long-established companies sometimes manage to restart above-average growth through, for example, innovation, expansion to new markets, or acquisitions. Or a company's long-run growth rate may be interrupted by a period of subnormal performance. If growth is expected to moderate (in the first case) or improve (in the second case) toward some long-term growth rate, a two-stage DDM may be appropriate. Thus, we chose a two-stage DDM to value International Business Machines Corporation in Example 10.

EXAMPLE 10

Two-Stage Dividend Discount Model: International Business Machines Corporation

International Business Machines Corporation (IBM) is a US based leading technology company. IBM was founded in 1911, initially as a company that manufactured machinery for sale and lease, ranging from commercial scales and industrial time recorders, meat and cheese slicers, to punched cards. IBM introduced the personal computer in 1981; however, by the 1990s, it began to suffer losses in its core computer manufacturing business and by the 2000s, it had begun to diversify into business consulting, which was finalized in 2005 when it sold its personal computer business to Chinese company Lenovo. IBM now operates through five segments: Cognitive Solutions, Global Business Services (GBS), Technology Services & Cloud Platforms, Systems, and Global Financing. The Cognitive Solutions segment delivers a spectrum of capabilities from descriptive, predictive, and prescriptive analytics to cognitive systems. Cognitive Solutions includes Watson, a cognitive computing platform that has the ability to interact in natural language, process big data, and learn from interactions with people and computers. The GBS segment provides clients with consulting, application management services, and global process services. The Technology Services & Cloud Platforms segment provides information technology infrastructure services. The Systems segment provides clients with infrastructure technologies. The Global Financing segment includes client financing, commercial financing, and remanufacturing and remarketing.

The 30 July 2018 *Value Line* report on IBM appears in Exhibit 5. IBM has increased its dividends every year over the past 15 years. Information from *Value Line* shows that the dividend growth is around 17.0 percent for the past 10 years, 13.5 percent for the past 5 years, and estimated to be 4.5 percent for 2015 to 2023. After a period of growth through acquisition and merger, the pattern suggests that IBM may be transitioning to a mature growth phase.

dividend growth to 2015–2023).

The CAPM can be used to estimate the required return, *r*, for IBM. The *Value Line* report (in the top left corner) estimates beta to be 0.95. Using the yield of about 2.0 percent on 10-year US Treasury notes as a proxy for the risk-free rate and assuming an equity risk premium of 5.0 percent, we find the estimate for *r* would be 6.75 percent [2.0% + 0.95(5.0%)].

To estimate the intrinsic value at the end of 2018, we use the 2018 dividend of US$6.21 from the *Value Line* report. The dividend is assumed to grow at a rate of 5.0 percent for one year and then 2.33 percent thereafter. The growth rate assumption for the first stage is consistent with the *Value Line* forecast for the 2018 and 2019 dividends. Our assumption of a 2.33 percent perpetual growth rate produces an 8-year growth rate assumption near 4.5%,[4] which is consistent with the *Value Line* forecast of 4.5 percent growth from 2015–2022. Thus:

D_{2019} = US$6.21(1 + 0.05) = US$6.5205

D_{2020} = US$6.21(1 + 0.05)(1 + 0.0233) = US$6.6724

D_{2021} = US$6.21(1 + 0.05)(1 + 0.0233)^2 = US$6.8279

$$V_{2020} = \frac{\text{US\$6.8279}}{0.0675 - 0.0233} = \text{US\$154.4774}$$

$$V_{2018} = \frac{\text{US\$6.5205}}{(1+0.0675)} + \frac{\text{US\$6.6724}}{(1+0.0675)^2} + \frac{\text{US\$154.4774}}{(1+0.0675)^2} \approx \text{US\$147.523}$$

Given a recent price of US$148.56, as noted at the top left corner of the *Value Line* report, the intrinsic-value estimate of US$147.523 suggests that IBM is approximately fairly valued.

Exhibit 5:

Value Line Report on IBM

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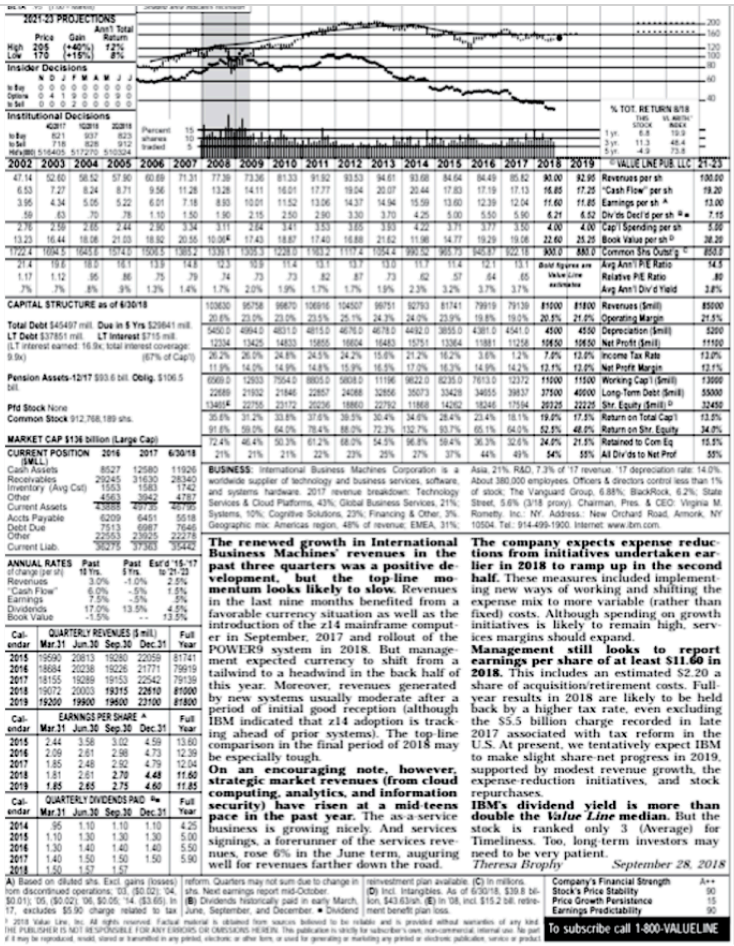

2021-23 PROJECTIONS

	Price	Gain	Ann'l Total Return
High	205	(+40%)	12%
Low	170	(+15%)	8%

Insider Decisions

	N	D	J	F	M	A	M	J	J
to Buy	0	0	0	0	0	0	0	0	0
Options	0	4	1	9	0	0	0	0	0
to Sell	0	0	0	2	0	0	0	0	0

Institutional Decisions

	4Q2017	1Q2018	2Q2018
to Buy	821	937	823
to Sell	718	828	912
Hld's(000)	518405	517270	510324

2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	© VALUE LINE PUB. LLC	21-23
47.14	52.60	58.52	57.90	60.69	71.31	77.39	73.36	81.33	91.92	93.53	94.61	93.68	84.64	84.49	85.82	90.00	92.95	Revenues per sh	100.00
6.53	7.27	8.24	8.71	9.56	11.26	13.28	14.11	16.01	17.77	19.04	20.07	20.44	17.83	17.19	17.13	16.85	17.25	"Cash Flow" per sh	19.20
3.95	4.34	5.05	5.22	6.01	7.18	8.93	10.01	11.52	13.06	14.37	14.94	15.59	13.60	12.39	12.04	11.60	11.80	Earnings per sh A	13.00
.59	.63	.70	.78	1.10	1.50	1.90	2.15	2.50	2.90	3.30	3.70	4.25	5.00	5.50	5.90	6.21	6.52	Div'ds Decl'd per sh B■	7.15
2.78	2.59	2.65	2.44	2.90	3.34	3.11	2.84	3.41	3.53	3.55	3.93	4.22	3.71	3.77	3.50	4.00	4.00	Cap'l Spending per sh	5.00
13.23	16.44	18.08	21.03	18.92	20.55	10.06[E]	17.43	18.87	17.40	16.88	21.62	11.98	14.77	19.29	19.08	22.60	25.25	Book Value per sh D	30.20
1722.4	1694.5	1643.6	1574.0	1506.5	1385.2	1339.1	1305.3	1228.0	1163.2	1111.4	1054.4	990.52	965.73	945.87	922.18	900.0	880.0	Common Shs Outst'g C	850.0
21.4	19.6	18.0	16.1	13.9	14.8	12.3	10.9	11.4	13.1	13.7	13.0	11.7	11.4	12.1	13.1	Bold figures are Value Line estimates		Avg Ann'l P/E Ratio	14.0
1.17	1.12	.96	.86	.75	.79	.74	.73	.73	.82	.87	.73	.62	.57	.64	.65			Relative P/E Ratio	.80
.7%	.7%	.8%	.9%	1.3%	1.4%	1.7%	2.0%	1.9%	1.7%	1.7%	1.9%	2.3%	3.2%	3.7%	3.7%			Avg Ann'l Div'd Yield	3.8%

CAPITAL STRUCTURE as of 6/30/18

Total Debt $45497 mill. Due in 5 Yrs $29841 mill.
LT Debt $37851 mill. LT Interest $715 mill.
(LT interest earned: 16.9x; total interest coverage: 9.9x) (67% of Cap'l)

Pension Assets-12/17 $93.6 bill. Oblig. $106.5 bill.

Pfd Stock None

Common Stock 912,768,189 shs.

MARKET CAP $136 billion (Large Cap)

	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019		21-23
	100630	95758	99870	106916	104507	99751	92793	81741	79919	79139	81000	81800	Revenues ($mill)	85000
	20.6%	23.0%	23.0%	23.5%	25.1%	24.3%	24.0%	23.9%	19.8%	19.0%	20.5%	21.0%	Operating Margin	21.5%
	5450.0	4994.0	4831.0	4815.0	4676.0	4678.0	4492.0	3855.0	4381.0	4541.0	4500	4550	Depreciation ($mill)	5200
	12334	13425	14833	15855	16604	16483	15751	13364	11881	11258	10650	10650	Net Profit ($mill)	11100
	26.2%	26.0%	24.8%	24.5%	24.2%	15.6%	21.2%	16.2%	3.6%	1.2%	7.0%	12.0%	Income Tax Rate	12.0%
	11.9%	14.0%	14.9%	14.8%	15.9%	16.5%	17.0%	16.3%	14.9%	14.2%	13.1%	13.0%	Net Profit Margin	13.1%
	6569.0	12933	7504.0	8805.0	5808.0	11196	9622.0	8235.0	7613.0	12372	11000	11500	Working Cap'l ($mill)	13000
	22689	21932	21846	22857	24388	32856	35073	33428	34655	39837	37500	40000	Long-Term Debt ($mill)	55000
	13465[E]	22755	23172	20236	18860	22792	11868	14262	18246	17594	20325	22225	Shr. Equity ($mill) D	32450
	35.6%	31.2%	33.8%	37.6%	39.5%	30.4%	34.6%	28.4%	23.4%	18.1%	19.0%	17.5%	Return on Total Cap'l	13.5%
	91.6%	59.0%	64.0%	78.4%	88.0%	72.3%	132.7%	93.7%	65.1%	64.0%	52.5%	48.0%	Return on Shr. Equity	34.0%
	72.4%	46.4%	50.3%	61.2%	68.0%	54.5%	96.8%	59.4%	36.3%	32.6%	24.0%	21.5%	Retained to Com Eq	15.5%
	21%	21%	21%	22%	23%	25%	27%	37%	44%	49%	54%	55%	All Div'ds to Net Prof	55%

CURRENT POSITION ($MILL.)	2016	2017	6/30/18
Cash Assets	8527	12580	11926
Receivables	29245	31630	28340
Inventory (Avg Cst)	1553	1583	1742
Other	4563	3942	4787
Current Assets	43888	49735	46795
Accts Payable	6209	6451	5518
Debt Due	7513	6987	7646
Other	22553	23925	22278
Current Liab.	36275	37363	35442

ANNUAL RATES of change (per sh)	Past 10 Yrs.	Past 5 Yrs.	Est'd '15-'17 to '21-'23
Revenues	3.0%	-1.0%	2.5%
"Cash Flow"	6.0%	-.5%	1.5%
Earnings	7.5%	-.5%	.5%
Dividends	17.0%	13.5%	4.5%
Book Value	-1.5%	--	13.5%

Cal-endar	QUARTERLY REVENUES ($ mill.) Mar.31	Jun.30	Sep.30	Dec.31	Full Year
2015	19590	20813	19280	22059	81741
2016	18684	20238	19226	21771	79919
2017	18155	19089	19153	22542	79139
2018	19072	20003	19315	22610	81000
2019	19200	19900	19600	23100	81800

Cal-endar	EARNINGS PER SHARE A Mar.31	Jun.30	Sep.30	Dec.31	Full Year
2015	2.44	3.58	3.02	4.59	13.60
2016	2.09	2.61	2.98	4.73	12.39
2017	1.85	2.48	2.92	4.79	12.04
2018	1.81	2.61	2.70	4.48	11.60
2019	1.85	2.65	2.75	4.60	11.85

Cal-endar	QUARTERLY DIVIDENDS PAID B■ Mar.31	Jun.30	Sep.30	Dec.31	Full Year
2014	.95	1.10	1.10	1.10	4.25
2015	1.10	1.30	1.30	1.30	5.00
2016	1.30	1.40	1.40	1.40	5.50
2017	1.40	1.50	1.50	1.50	5.90
2018	1.50	1.57	1.57	--	

% TOT. RETURN 6/18

	THIS STOCK	VL ARITH. INDEX
1 yr.	6.8	19.9
3 yr.	11.3	48.4
5 yr.	4.9	73.8

BUSINESS: International Business Machines Corporation is a worldwide supplier of technology and business services, software, and systems hardware. 2017 revenue breakdown: Technology Services & Cloud Platforms, 43%; Global Business Services, 21%; Systems, 10%; Cognitive Solutions, 23%; Financing & Other, 3%. Geographic mix: Americas region, 48% of revenue; EMEA, 31%; Asia, 21%. R&D, 7.3% of '17 revenue. '17 depreciation rate: 14.0%. About 380,000 employees. Officers & directors control less than 1% of stock. The Vanguard Group, 6.88%; BlackRock, 6.2%; State Street, 5.6% (3/18 proxy). Chairman, Pres. & CEO: Virginia M. Rometty. Inc.: NY. Address.: New Orchard Road, Armonk, NY 10504. Tel.: 914-499-1900. Internet: www.ibm.com.

The renewed growth in International Business Machines' revenues in the past three quarters was a positive development, but the top-line momentum looks likely to slow. Revenues in the last nine months benefited from a favorable currency situation as well as the introduction of the z14 mainframe computer in September, 2017 and rollout of the POWER9 system in 2018. But management expected currency to shift from a tailwind to a headwind in the back half of this year. Moreover, revenues generated by new systems usually moderate after a period of initial good reception (although IBM indicated that z14 adoption is tracking ahead of prior systems). The top-line comparison in the final period of 2018 may be especially tough.

On an encouraging note, however, strategic market revenues (from cloud computing, analytics, and information security) have risen at a mid-teens pace in the past year. The as-a-service business is growing nicely. And services signings, a forerunner of the services revenues, rose 6% in the June term, auguring well for revenues farther down the road.

The company expects expense reductions from initiatives undertaken earlier in 2018 to ramp up in the second half. These measures included implementing new ways of working and shifting the expense mix to more variable (rather than fixed) costs. Although spending on growth initiatives is likely to remain high, services margins should expand.

Management still looks to report earnings per share of at least $11.60 in 2018. This includes an estimated $2.20 a share of acquisition/retirement costs. Full-year results in 2018 are likely to be held back by a higher tax rate, even excluding the $5.5 billion charge recorded in late 2017 associated with tax reform in the U.S. At present, we tentatively expect IBM to make slight share-net progress in 2019, supported by modest revenue growth, the expense-reduction initiatives, and stock repurchases.

IBM's dividend yield is more than double the *Value Line* median. But the stock is ranked only 3 (Average) for Timeliness. Too, long-term investors may need to be very patient.

Theresa Brophy *September 28, 2018*

A) Based on diluted shs. Excl. gains (losses) from discontinued operations: '03, ($0.02); '04, $0.01); '05, ($0.02); '06, $0.05; '14, ($3.65). In '17, excludes $5.90 charge related to tax reform. Quarters may not sum due to change in shs. Next earnings report mid-October. B) Dividends historically paid in early March, June, September, and December. ■ Dividend reinvestment plan available. (C) In millions. (D) Incl. Intangibles. As of 6/30/18, $39.8 billion, $43.63/sh. (E) In '08, incl. $15.2 bill. retirement benefit plan loss.

Company's Financial Strength	A++
Stock's Price Stability	90
Price Growth Persistence	15
Earnings Predictability	90

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LB

In the IBM example, why do we compute D_2021? It is assumed that the dividend grows at 5% rate in the first year and then grows at the sustainable growth rate of 2.33% from the second year.

In any case, I think there are some errors in the compounding growth rate in D_2020 and D_2021, as it should be D_2020=D_2019*(1+2,33%).

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